



1.7c House Flooring Project

You have been sought out by the famous television network, HGTV, to provide an estimate for the construction crew with the renovation of a family's house. After many years of having carpet throughout the house, the family has decided they want their entire house to be water-resistant laminate with a decorative quarter-round trim.

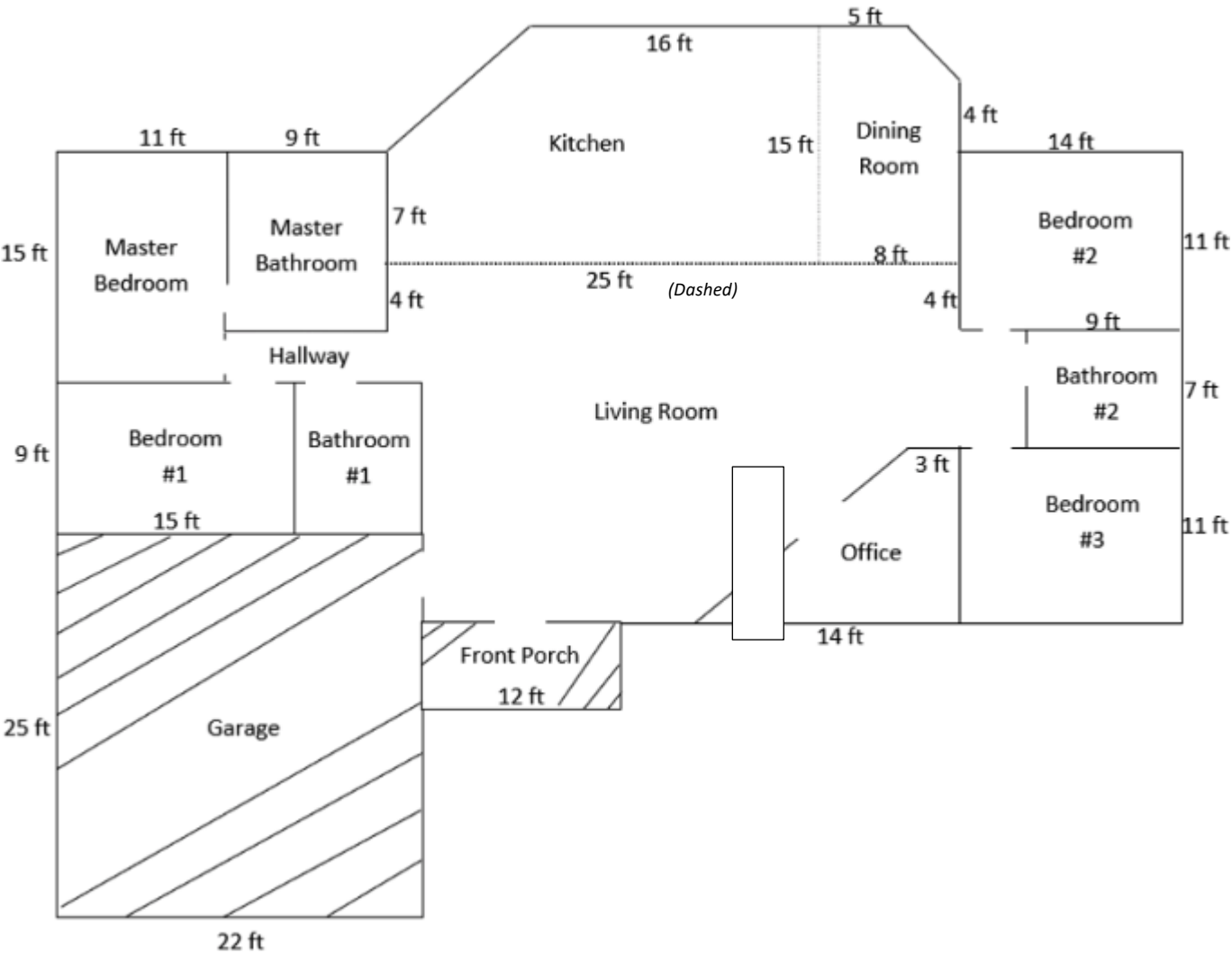
Your task is to find the total price of the new flooring. There are three important pieces of material needed for the new floors: 1) the underlayment. This is the stuff that goes beneath the floor to provide a smooth, even surface and cushions the floor so it is not too hard; 2) the laminate flooring itself. This is what you walk on. It often looks like fake wood; and 3) the trim. This is what goes along the space where the walls meet the floor and make things look nice. You will need to calculate the perimeter and area of the rooms throughout the house and use the information about the products below to figure out the total cost.

Note: You do not need to provide trim on the bottom of door frames!

Points Earned:	0	2	4	6
Rubric	You did not complete the assignment, cheated on it, or did not demonstrate understanding of how to calculate the perimeter and area of the shapes covered in class.	You correctly calculated the perimeter and area of <i>some</i> of the shapes involved in the house, but did not provide an estimate.	You correctly calculated the perimeter and area of <i>nearly all</i> the shapes involved, but did not provide an estimate.	You correctly calculated the perimeter and area of <i>nearly all</i> the shapes involved and provided a correct estimate.

House Floor Plan

Assume right angles exist where there appear to be right angles. All door frames are 3ft wide.



Part 1: Complete the following chart by using the dimensions of each room from the floor plan. Show your work for each calculation.

Room	Work (mathematics) showing your Area/Perimeter Calculations		Area of the room (ft ²) =	Perimeter needed (ft) =
	Area	Perimeter		
Bath #1				
Bath #2				
Bedroom #1				
Bedroom #2				
Bedroom #3				

Dining Room				
Kitchen				
Living Room + hallway + in front of bathroom #2				
Master Bath				
Master Bed				
Office				
Total				

Part 2: Use the costs for the materials on the next page to provide an invoice. Show all work.

Total cost for underlayment. Show all calculations. *Note: can you buy a part of a roll of underlayment from the store?*

Total cost for laminate flooring. Show all calculations. *Note: can you buy a partial box of flooring from the store?*

Total cost for trim. Show all calculations. *Note: can you buy a piece of from the store?*

Total Price for Flooring Project:

Cost of Materials

Underlayment: AirGuard 5-in-1 Underlayment with Microban sold as a roll.

- 40 in. x 30 ft. x 2 mm
- 100 sq. ft. per roll
- \$53.21 per roll



Laminate Flooring: Water Resistant EIR Silverton Oak Laminate Flooring sold in cases.

- 8 mm Thickness
- 7-1/2 in. Wide x 50-2/3 in. Length
- 23.69 sq. ft./ case
- Price: \$44.77/ case



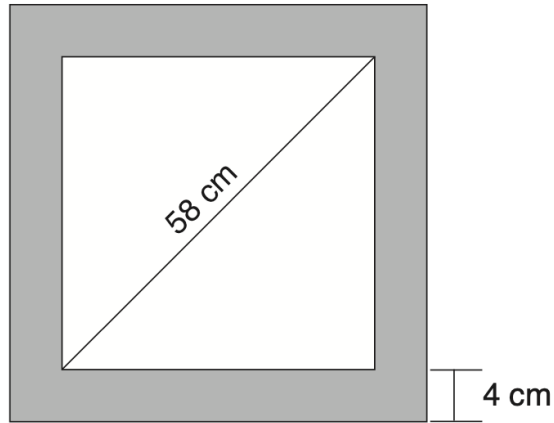
Trim: White Laminate Quarter Round Molding sold in individual strips.

- 3/4 in. Thick x 3/4 in. Wide
- 94-1/4 in. Length
- \$8.48 per unit



Optional Challenge:

Keira has a square poster that she is framing and placing on her wall. The poster has a diagonal 58 cm long and fits exactly inside the frame. The width of the frame around the picture is 4 cm.



Determine and state the total area of the poster and frame to the *nearest tenth of a square centimeter*.